

Solution Requirements

A Comprehensive Measure of Well-Being | HDI Predictor

Functional Requirements

- FR1: Accept life expectancy (20–85 years)
- FR2: Accept mean years of schooling (0–15)
- FR3: Accept expected years of schooling (0–18)
- FR4: Accept GNI per capita (100–75000 USD PPP)
- FR5: Compute HDI using UNDP geometric mean formula
- FR6: Classify result into Very High / High / Medium / Low
- FR7: Provide 20 country presets from UNDP HDR 2023/24

Non-Functional Requirements

- NFR1: Response time < 100ms (client-side computation)
- NFR2: Mobile responsive design
- NFR3: Accuracy within ± 0.005 of official UNDP scores
- NFR4: 99.9% uptime via Vercel deployment

Constraints

- Must use official UNDP formula (not arbitrary ML model)
- Frontend-only computation (no backend server required)